

essential characteristics — on demand service, access network, resource pooling, elasticity and measured services. To achieve these, cloud computing provides three kinds of service models: Software as a Service (SaaS), Platform as a Service (PaaS) and Infrastructure as a Service (IaaS).

The key business drivers of PaaS adoption are:

- Reduction of capex and opex to deliver business services
- Minimising IT costs by improving the delivery time and quality of the application development and delivery
- Increasing the flexibility and integration between middleware components

Simple PaaS is the entry point into the PaaS space. It allows provisioning of application services and exposes them into a self-service catalogue; it automates the deployment and meters the resources used by this service.

Manage PaaS manages the SLA and QoS aspects of the provisioned applications such as resiliency, application performance, security, etc.

Programming PaaS allows applications to integrate with external applications or public clouds, and to implement auto-scaling and cloud-bursting scenarios.

Process-oriented PaaS allows implementation of a DevOps process by creating a continuous delivery flow that automates the build, test and delivery of applications into a cloud environment.

In addition to these adoption patterns, there are other variations of PaaS, as listed below. These variations might align to one of the patterns explained above.

iPaaS: Integration Platform as a Service (iPaaS) is a suite of cloud services that enables development, execution and governance of integration flows connecting any combination of on-premises and cloud-based

processes, services, applications and data within individuals or across multiple organisations. Examples are MuleSoft CloudHub and BizTalk.

mPaaS: Mobile Platform as a Service (mPaaS) is a provision of an interactive development environment (IDE) for the creation of mobile apps. It supports multiple mobile operating platforms.

dbPaaS: Database Platform as a Service (dbPaaS) is an on-demand, secure and scalable self-service database platform that automates the provisioning and administration of databases. dbPaaS makes it easier to scale databases and makes them more reliable.

IoT PaaS: This provides common infrastructure to enable communication, security, analytics and management for heterogeneous IoT topologies. It provides simpler and agile models for building IoT solutions.

bpmPaaS: Business process management PaaS (bpmPaaS) is a complete pre-integrated BPM platform hosted in the cloud and delivered as a service. It is leveraged for the development and execution of business processes and workflow-centric applications across enterprises. Examples are Pega cloud, and OpenText Cordys cloud.

Some basic characteristics of PaaS are:

- Services to develop, test, deploy, host and maintain applications in the same integrated development environment
- Multi-tenant architecture, in which multiple concurrent users use the same development application
- Built-in scalability of deployed software, including load balancing and failover
- Integration with heterogeneous platforms and systems
- Support for development team collaboration
- Tools to handle billing and subscription management

Key open source Platforms as a Service

Before choosing a PaaS, enterprises must consider the following:

- Deployment flexibility
- Ease of operations
- Choice of application stacks
- Language, database and framework support
- Scaling capabilities
- QoS
- Tooling for development and operations
- How well it fits your business

Let's now take a quick look at some popular open source PaaS.

Cloud Foundry: This PaaS provides a choice of clouds, developer frameworks and application services. Cloud Foundry makes it faster and easier to build, test, deploy and scale applications.

It has different distributions, of which the popular ones are Pivotal and IBM. It contains application runtime and container runtime. It also has Pivotal application service and Pivotal container service.

OpenShift: This is Red Hat's cloud computing PaaS offering. It is an application platform in the cloud, where application developers and teams can build, test, deploy and run their applications.

Cloudify: Cloudify was developed and designed on the principles of openness to power the IT transformation revolution. It enables organisations to design, build and deliver various business applications and network services. The latest version of Cloudify is 4.3, which incorporates enhanced features like advanced security, control and true self-service. Cloudify 4.3 introduced a totally new concept for container orchestration with Kubernetes.

Table 1 lists the basic functionality and its corresponding Cloud Foundry, Cloudify and OpenShift architectural components. This is purely based on my views and the authenticity of the